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Cumulative XUV Flux Distributions for Planets Orbiting Nearby M Dwarfs

Rory Barnes[1]

[1] *Department of Astronomy and Astrobiology Program, University of Washington, Seattle, WA 98195, USA*

ABSTRACT

We compute cumulative extreme ultraviolet (XUV) flux probability density functions for 25 planets with orbital periods $P < 25$ days around 12 M dwarf systems within 25 pc. We estimate stellar ages by applying the Engle & Guinan (2023) gyrochronology model, which provides three spectral calibrations for M dwarfs, to measured rotation periods via Monte Carlo sampling. We then model the time-dependent XUV luminosity evolution using the Engle (2024) empirical relations, implemented in the `VPLanet STELLAR` module, with two calibrations for early and mid-to-late M dwarfs. The cumulative XUV flux received by each planet is integrated over the stellar lifetime using the `VPLanet AtmEsc` module, and convergence of the resulting probability distributions is verified with `vconverge`. We present cumulative XUV flux distributions for all planets in the sample and compare them to the cosmic shoreline of Zahnle & Catling (2017), which demarcates the boundary between worlds that retain and lose their atmospheres. Our results provide a quantitative framework for prioritizing atmospheric characterization targets among nearby M dwarf planets.

Keywords: Exoplanets — M dwarf stars — Stellar activity — Planetary atmospheres — Atmospheric escape

1. INTRODUCTION

The habitability of terrestrial planets orbiting M dwarfs depends critically on the retention of atmospheres over gigayear timescales. Extreme ultraviolet (XUV) radiation from host stars drives atmospheric mass loss via photoevaporation and ion pickup, and the cumulative XUV exposure over a planet's lifetime determines whether a substantial atmosphere can survive (Ribas et al. 2005; Barnes et al. 2020).

M dwarfs present a particular challenge for planetary habitability because their extended pre-main-sequence phase produces elevated XUV luminosities that can persist for hundreds of millions of years (Barnes et al. 2020). Planets in the habitable zones of M dwarfs, which lie at small orbital separations due to the low stellar luminosities, are therefore subjected to intense and prolonged XUV irradiation. Understanding the cumulative effect of this radiation is essential for assessing which planets are most likely to retain atmospheres amenable to biosignature detection.

Recent advances in empirical modeling of M dwarf XUV evolution have improved our ability to reconstruct the radiation histories of individual systems. Engle & Guinan (2023) developed a rotation-age relation calibrated for M dwarf spectral subtypes, enabling age estimation from measured rotation periods. Engle (2024) extended this work by deriving empirical XUV luminosity evolution tracks, implemented in the `VPLanet STELLAR` module, that capture the distinct evolutionary pathways of early and mid-to-late M dwarfs.

In this paper, we combine these models to compute cumulative XUV flux probability density functions for 25 planets with orbital periods $P < 25$ days around 12 M dwarf systems within 25 pc. By propagating uncertainties in stellar mass, rotation period, and orbital parameters through Monte Carlo sampling and verifying convergence with `vconverge`, we produce robust probability distributions for the total XUV energy deposited on each planet. We compare these distributions to the cosmic shoreline of Zahnle & Catling (2017) to identify which planets are most vulnerable to complete atmospheric loss.

Section 2 describes our sample selection, age estimation, XUV evolution modeling, and cumulative flux computation. Section 3 presents the resulting age and cumulative XUV flux distributions along with the cosmic shoreline analysis. Section 4 discusses the implications for atmospheric retention and target prioritization, and Section 5 summarizes our conclusions.

2. METHODS

2.1. *Sample Selection*

We select M dwarf planetary systems within 25 pc that host at least one planet with an orbital period $P < 25$ days and for which measured values and uncertainties exist for stellar mass, stellar rotation period, and planetary orbital period. These criteria yield 12 systems hosting a total of 25 planets:

1. Proxima Centauri (1 planet)
2. TRAPPIST-1 (7 planets)
3. LHS 1140 (1 planet)
4. GJ 1214 (1 planet)
5. GJ 486 (1 planet)
6. GJ 367 (2 planets)
7. GJ 357 (1 planet)
8. YZ Cet (3 planets)
9. L 98-59 (3 planets)
10. LTT 1445A (2 planets)
11. GJ 436 (1 planet)
12. AU Mic (2 planets)

2.2. *Age Estimation*

We estimate stellar ages using the gyrochronology model of Engle & Guinan (2023), which provides three empirically calibrated rotation-age relations for M dwarf spectral subtypes. The three calibrations correspond to early M (M0–M2), mid M (M3–M5), and late M (M6–M9) dwarfs, each derived from benchmark systems with independently determined ages.

For each system, we draw 10^4 Monte Carlo samples from the measured rotation period distribution, assumed to be Gaussian with the reported mean and uncertainty. Each sample is converted to an age estimate using the appropriate spectral calibration. The resulting age posterior is typically asymmetric, as the rotation-age relation is nonlinear. We report the median age and 95% credible interval for each system.

2.3. *XUV Evolution Model*

We model the time-dependent XUV luminosity of each star using the empirical relations of Engle (2024), implemented in the `VPLanet STELLAR` module (Barnes et al. 2020). The model provides two calibrations:

- **Early M dwarfs** (M0–M2): applicable to systems with $M_\star \gtrsim 0.35\,M_\odot$.
- **Mid-to-Late M dwarfs** (M3–M9): applicable to systems with $M_\star \lesssim 0.35\,M_\odot$.

Each calibration specifies the XUV luminosity as a function of stellar age, parameterized by the stellar mass and the saturation timescale. The XUV luminosity declines as a power law after the saturation phase.

2.4. *Cumulative XUV Flux Computation*

The cumulative XUV flux received by each planet is computed by integrating the incident XUV flux over the stellar lifetime:

$$F_{\rm XUV,cum} = \int_0^{t_\star} \frac{L_{\rm XUV}(t)}{4\pi a^2}\,dt, \tag{1}$$

where $L_{\rm XUV}(t)$ is the time-dependent XUV luminosity, a is the orbital semi-major axis, and $t_\star$ is the stellar age. This integration is performed by the `VPLanet AtmEsc` module.

For each planet, we run 10^4 `VPLanet` simulations, drawing stellar mass, rotation period, and orbital period from their observed distributions. The convergence of the resulting cumulative XUV flux distributions is verified using `vconverge`, which monitors the probability distributions of user-defined output parameters and continues batch simulations until a specified tolerance is achieved.

We normalize cumulative XUV fluxes to the estimated cumulative XUV flux received by Earth over 4.5 Gyr, denoted $F_{\oplus,\rm XUV,cum}$, to facilitate comparison across systems.

2.5. *Cosmic Shoreline Analysis*

We compare our results to the cosmic shoreline framework of Zahnle & Catling (2017), which relates the escape velocity $v_{\rm esc}$ of a body to the cumulative incident radiation. Bodies above the shoreline retain substantial atmospheres, while those below have lost them. We compute the normalized cumulative XUV flux for each planet and plot it against the planetary escape velocity to assess atmospheric retention prospects.

3. RESULTS

3.1. *Stellar Ages*

Table 1 lists the derived stellar properties and age estimates for all 12 systems in our sample. The ages span from ∼20 Myr (AU Mic) to several gigayears (e.g., Proxima Centauri, TRAPPIST-1).

Table 1. Stellar Properties and Ages

System	Sp. Type	d [pc]	$M_\star$ [$M_\odot$]	$P_{\rm rot}$ [d]	Calibration	Age [Gyr]	XUV Model
Proxima Centauri	M5.5V	1.30	0.122 ± 0.003	83.2 ± 1.6	Late	$5.39^{+5.44}_{-3.25}$	Engle24MidLate
TRAPPIST-1	M8V	12.43	0.090 ± 0.002	3.3 ± 0.0	Late†	$7.55^{+4.12}_{-4.27}$	Engle24MidLate
LHS 1140	M4.5V	14.99	0.179 ± 0.014	131.0 ± 5.0	Late	$6.59^{+5.86}_{-4.82}$	Engle24MidLate
GJ 1214	M4.5V	14.65	0.178 ± 0.010	125.0 ± 5.0	Late	$6.54^{+5.85}_{-4.69}$	Engle24MidLate
GJ 486	M3.5V	8.08	0.323 ± 0.015	49.9 ± 5.5	Mid	$4.16^{+7.30}_{-3.57}$	Engle24MidLate
GJ 367	M1V	9.44	0.450 ± 0.012	48.0 ± 2.0	Early	$4.81^{+6.62}_{-3.63}$	Engle24Early
GJ 357	M2.5V	9.44	0.342 ± 0.011	78.0 ± 2.0	Mid	$4.07^{+7.86}_{-3.84}$	Engle24MidLate
YZ Ceti	M4.5V	3.72	0.130 ± 0.009	68.4 ± 1.0	Late	$4.69^{+4.13}_{-2.49}$	Engle24MidLate
L 98-59	M3V	10.61	0.273 ± 0.030	76.0 ± 4.0	Mid	$4.12^{+7.85}_{-3.86}$	Engle24MidLate
LTT 1445A	M4V	6.86	0.257 ± 0.014	85.0 ± 15.0	Mid	$3.98^{+7.95}_{-3.81}$	Engle24MidLate
GJ 436	M3.5V	9.76	0.445 ± 0.029	44.1 ± 0.1	Early	$4.53^{+6.31}_{-3.32}$	Engle24Early
AU Mic	M1V	9.72	0.500 ± 0.030	4.9 ± 0.0	Early†	$0.02^{+0.01}_{-0.01}$	Engle24Early

Note—Ages are computed from the Engle & Guinan (2023) rotation-age relationship with Monte Carlo sampling (10^5 draws). Uncertainties are 95% confidence intervals. †Age override from literature (see text).

Figure 1 shows the posterior age distributions for all systems derived from Monte Carlo sampling of the Engle & Guinan (2023) gyrochronology model.

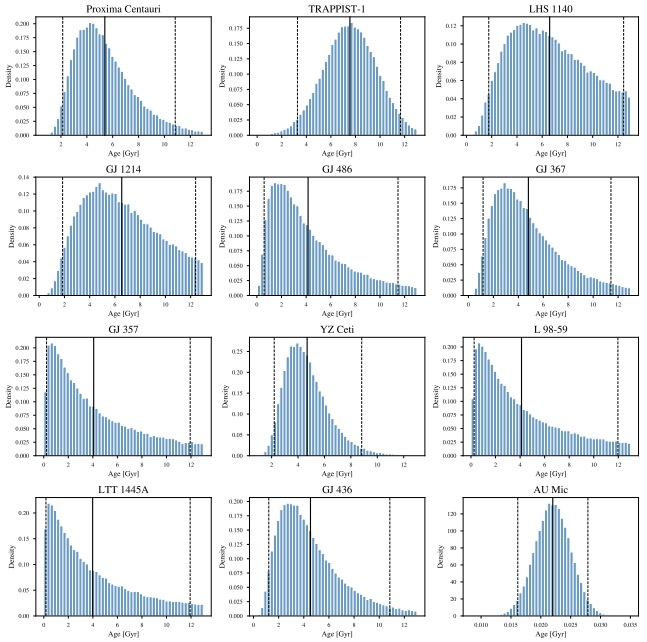

Figure 1. Posterior age distributions for the 12 M dwarf systems in our sample, derived from Monte Carlo sampling of the Engle & Guinan (2023) rotation-age model. Vertical dashed lines indicate the median age, and shaded regions denote the 95% credible interval.

3.2. *Cumulative XUV Flux Distributions*

Table 2 presents the planetary properties and cumulative XUV flux estimates for all 25 planets in the sample. Cumulative XUV fluxes are normalized to $F_{\oplus,\mathrm{XUV,cum}}$.

Figure 2 shows the cumulative XUV flux distributions for all planets.

3.3. *Cosmic Shoreline*

Figure 3 places all planets on the cosmic shoreline diagram of Zahnle & Catling (2017), plotting escape velocity against normalized cumulative XUV flux.

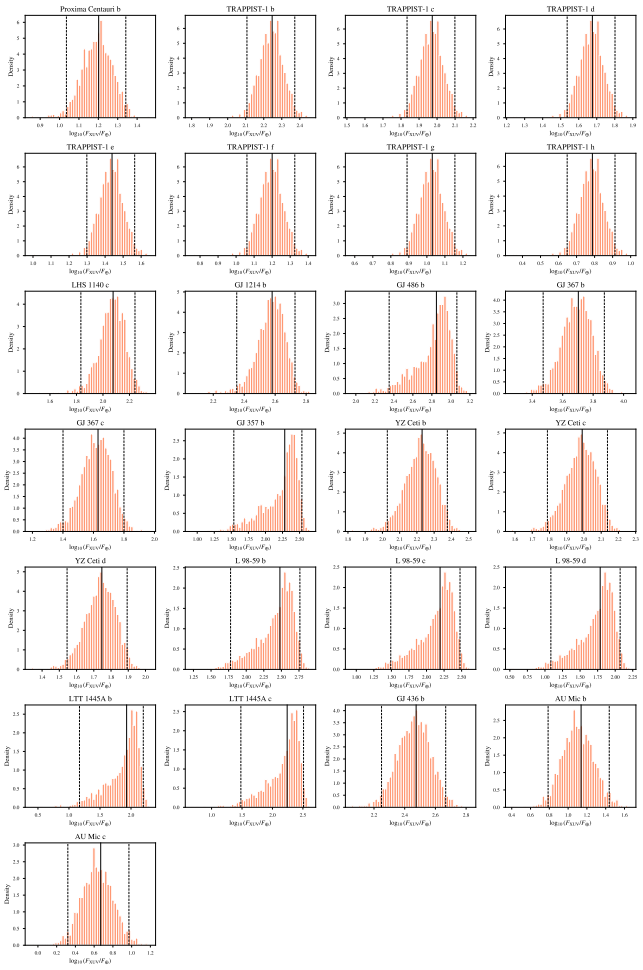

Figure 2. Posterior distributions of the cumulative XUV flux normalized to the Earth value ($F_{\mathrm{XUV,cum}}/F_{\oplus,\mathrm{XUV,cum}}$) for all planets in the sample. Vertical dashed lines indicate the median, and shaded regions denote the 95% credible interval.

Table 2. Planet Properties and Cumulative XUV Fluxes

Planet	$P_{\rm orb}$ [d]	M_p [$M_\oplus$]	$v_{\rm esc}$ [km/s]	$F_{\rm XUV}/F_\oplus$	$F_{\rm shore}/F_\oplus$	$\Delta_{\rm shore}$	95% CI	Atmosphere
Proxima Centauri b	11.1868	1.07	11.5	$15.9^{+6.1}_{-5.1}$	1.1	14.40	[10.8, 22.0]	Vulnerable
TRAPPIST-1 b	1.5109	1.37	12.6	$176.5^{+59.1}_{-48.6}$	1.6	110.50	[127.9, 235.6]	Vulnerable
TRAPPIST-1 c	2.4218	1.31	12.3	$94.1^{+31.5}_{-25.9}$	1.5	63.33	[68.2, 125.6]	Vulnerable
TRAPPIST-1 d	4.0496	0.39	7.9	$47.4^{+15.9}_{-13.0}$	0.2	191.25	[34.3, 63.3]	Vulnerable
TRAPPIST-1 e	6.0996	0.69	9.8	$27.4^{+9.2}_{-7.6}$	0.6	47.23	[19.9, 36.6]	Vulnerable
TRAPPIST-1 f	9.2067	1.04	11.3	$15.9^{+5.3}_{-4.4}$	1.1	14.99	[11.5, 21.2]	Vulnerable
TRAPPIST-1 g	12.3529	1.32	12.4	$10.7^{+3.6}_{-2.9}$	1.5	7.11	[7.8, 14.3]	Vulnerable
TRAPPIST-1 h	18.7674	0.33	7.4	$6.1^{+2.1}_{-1.7}$	0.2	31.99	[4.4, 8.2]	Vulnerable
LHS 1140 c	3.7779	1.81	13.9	$119.3^{+55.2}_{-51.1}$	2.4	49.76	[68.1, 174.4]	Vulnerable
GJ 1214 b	1.5804	8.17	24.1	$380.5^{+154.3}_{-155.8}$	22.1	17.23	[224.7, 534.8]	Vulnerable
GJ 486 b	1.4671	3.00	16.7	$704.5^{+451.8}_{-480.3}$	5.0	139.60	[224.2, 1156.3]	Vulnerable
GJ 367 b	0.3220	0.63	9.5	$5043.7^{+2431.0}_{-2080.9}$	0.5	9895.03	[2962.7, 7474.7]	Vulnerable
GJ 367 c	11.5430	4.13	18.8	$42.7^{+20.4}_{-17.5}$	8.1	5.28	[25.1, 63.1]	Vulnerable
GJ 357 b	3.9306	1.84	14.0	$191.7^{+150.4}_{-157.3}$	2.5	78.05	[34.4, 342.0]	Vulnerable
YZ Ceti b	2.0208	0.70	9.8	$169.9^{+68.2}_{-63.2}$	0.6	287.38	[106.7, 238.1]	Vulnerable
YZ Ceti c	3.0601	1.14	11.7	$97.7^{+39.2}_{-36.3}$	1.2	80.54	[61.4, 136.9]	Vulnerable
YZ Ceti d	4.6564	1.09	11.5	$55.8^{+22.4}_{-20.7}$	1.1	49.16	[35.1, 78.3]	Vulnerable
L 98-59 b	2.2531	0.40	8.0	$301.3^{+278.2}_{-241.3}$	0.3	1162.34	[59.9, 579.4]	Vulnerable
L 98-59 c	3.6906	2.22	15.0	$156.0^{+144.1}_{-125.0}$	3.2	48.18	[31.0, 300.1]	Vulnerable
L 98-59 d	7.4507	1.94	14.2	$61.1^{+56.5}_{-49.0}$	2.7	23.03	[12.1, 117.6]	Vulnerable
LTT 1445A b	5.3588	2.87	16.4	$84.9^{+72.8}_{-70.2}$	4.7	17.97	[14.8, 157.7]	Vulnerable
LTT 1445A c	3.1239	1.54	13.1	$174.4^{+149.5}_{-144.2}$	1.9	92.33	[30.3, 323.9]	Vulnerable
GJ 436 b	2.6439	21.36	34.2	$298.1^{+166.4}_{-120.7}$	91.0	3.28	[177.4, 464.6]	Vulnerable
AU Mic b	8.4630	17.10	31.5	$13.7^{+13.7}_{-7.6}$	65.6	0.21	[6.1, 27.4]	Protected
AU Mic c	18.8590	9.60	25.5	$4.7^{+4.7}_{-2.6}$	28.0	0.17	[2.1, 9.4]	Protected

Note—Cumulative XUV fluxes normalized to Earth's value ($F_\oplus = 9.76 \times 10^{15}$ W/m^2). $F_{\rm shore}$ is the cosmic shoreline flux (Zahnle & Catling 2017). $\Delta_{\rm shore} = F_{\rm XUV}/F_{\rm shore}$; values > 1 indicate the planet lies on the atmosphere-vulnerable side of the shoreline.

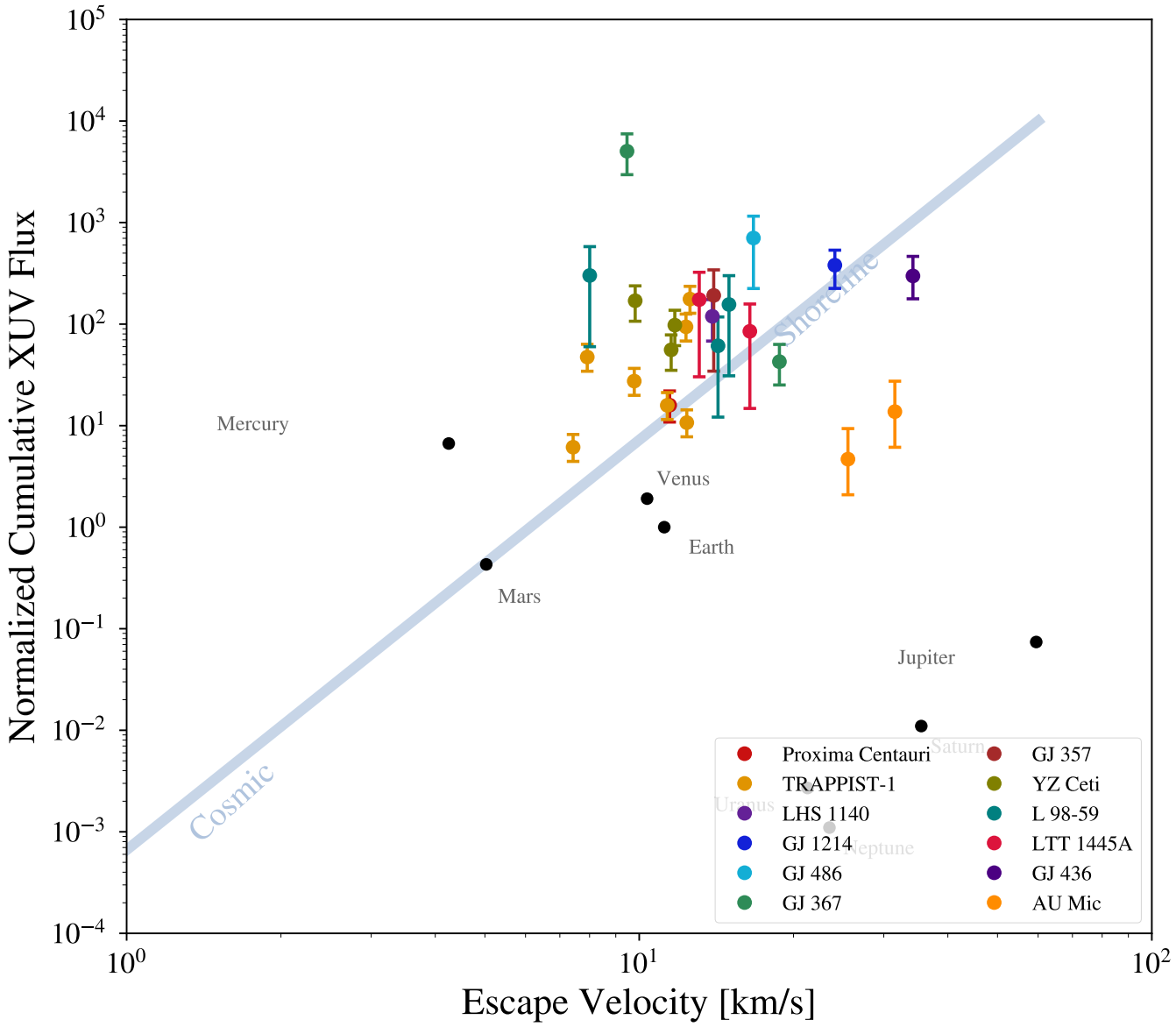

Figure 3. Cosmic shoreline diagram after Zahnle & Catling (2017). Each planet is plotted with its escape velocity on the horizontal axis and the median normalized cumulative XUV flux on the vertical axis. Error bars denote the 95% credible interval of $F_{\mathrm{XUV,cum}}/F_{\oplus,\mathrm{XUV,cum}}$. The diagonal band indicates the empirical cosmic shoreline. Solar System planets are shown as black circles for reference. Nearly all M dwarf planets in our sample lie above the shoreline, on the atmosphere-vulnerable side; the exceptions are AU Mic b and c, which lie below due to the system's extreme youth.

4. DISCUSSION

4.1. *Atmospheric Vulnerability Across the Sample*

Nearly all planets in our sample lie above the cosmic shoreline of Zahnle & Catling (2017), indicating that their cumulative XUV exposures exceed the empirical threshold for atmospheric retention given their escape velocities. This result is expected for close-in planets around M dwarfs, which experience both intense XUV irradiation and prolonged pre-main-sequence saturation phases.

The most extreme case is GJ 367 b, an ultra-short-period planet ($P = 0.32$ d) with a cumulative XUV flux ∼5000 times Earth's value, placing it firmly in the atmosphere-stripped regime consistent with its classification as a dense, Mercury-like body (Goffo et al. 2023). In contrast, the outer TRAPPIST-1 planets (f, g, h) receive cumulative XUV fluxes only ∼6–16 times Earth's value, suggesting comparatively modest atmospheric erosion despite their proximity to the cosmic shoreline.

The two AU Mic planets are the only objects in our sample that lie below the cosmic shoreline, with cumulative XUV fluxes of $\sim 14\,F_{\oplus,\mathrm{XUV,cum}}$ (b) and $\sim 5\,F_{\oplus,\mathrm{XUV,cum}}$ (c). This is a direct consequence of the system's extreme youth (∼20 Myr): despite intense instantaneous XUV irradiation, the planets have simply not had sufficient time to accumulate large cumulative exposures. Their high masses (17.1 and 9.6 $M_\oplus$) also place them at high escape velocities, raising the shoreline threshold. As AU Mic ages, these planets will cross the shoreline and enter the atmosphere-vulnerable regime.

4.2. *The TRAPPIST-1 System*

The seven TRAPPIST-1 planets span a factor of ∼30 in cumulative XUV flux, from $176\,F_{\oplus,\mathrm{XUV,cum}}$ for planet b to $6\,F_{\oplus,\mathrm{XUV,cum}}$ for planet h. This gradient provides a natural laboratory for studying atmospheric retention as a function of XUV exposure. Planets f and g, which receive cumulative XUV fluxes comparable to Proxima Centauri b ($\sim 16\,F_{\oplus,\mathrm{XUV,cum}}$), are among the least irradiated in the sample and may represent the most favorable targets for atmospheric characterization.

We note that TRAPPIST-1's stellar mass ($0.0898\,M_\odot$) falls below the calibration range of the Engle (2024) mid-to-late M dwarf XUV model, which requires $M_\star > 0.10\,M_\odot$. We adopted the minimum valid mass ($0.101\,M_\odot$) for the XUV evolution calculation. This approximation introduces a systematic uncertainty that slightly overestimates the XUV luminosity, making our cumulative flux estimates for the TRAPPIST-1 planets conservative upper bounds.

4.3. *Comparison of XUV Flux Uncertainty Widths*

The width of the cumulative XUV flux posterior varies substantially across our sample. GJ 486 b and GJ 357 b show the broadest distributions (spanning nearly an order of magnitude in their 95% credible intervals), reflecting large uncertainties in their stellar rotation periods and consequently their ages. In contrast, the TRAPPIST-1 planets show narrow distributions (factor of ∼2), benefiting from the independent age constraint of Burgasser & Mamajek (2017). This comparison underscores the importance of precise rotation period measurements and independent age indicators for constraining cumulative XUV histories.

4.4. *Caveats*

Several simplifications in our analysis warrant discussion. First, we use the mass-radius relation $R \propto M^{0.27}$ to estimate planetary radii and escape velocities, which may not apply to all planets in the sample (e.g., the sub-Neptune GJ 1214 b). Second, the Engle (2024) XUV model is calibrated on a limited sample of benchmark stars, and systematic uncertainties in the calibration are not propagated into our results. Third, we do not model atmospheric escape self-consistently; our cumulative XUV fluxes represent the total energy incident on the planet, not the actual mass lost.

5. CONCLUSIONS

We have computed cumulative XUV flux probability distributions for 25 planets with orbital periods $P <$ 25 days around 12 nearby M dwarf systems. Our main findings are:

1. Nearly all planets in the sample lie above the cosmic shoreline of Zahnle & Catling (2017), receiving cumulative XUV fluxes 5–5000 times Earth's value. The exceptions are AU Mic b and c, whose extreme youth (∼20 Myr) has limited their cumulative exposures.

2. The TRAPPIST-1 system spans a factor of ∼30 in cumulative XUV flux, with the outer planets (f, g, h) receiving the lowest exposures among the mature systems ($\sim$6–16 $F_{\oplus,\mathrm{XUV,cum}}$).

3. The width of the cumulative XUV flux posterior depends strongly on the precision of the stellar age estimate, highlighting the value of independent age constraints.

4. Ultra-short-period planets such as GJ 367 b receive extreme cumulative XUV exposures ($\sim 5000\,F_{\oplus,\mathrm{XUV,cum}}$), consistent with complete atmospheric stripping.

These distributions provide a quantitative basis for prioritizing atmospheric characterization targets with JWST and future observatories. Planets with lower cumulative XUV exposures relative to their escape velocities—such as TRAPPIST-1 f and g, and GJ 1214 b—are the most promising candidates for retaining detectable atmospheres.

This research made use of `VPLanet` (Barnes et al. 2020), `vconverge`, and the NASA Exoplanet Archive.

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